

Refugee Inflows and Native Parity Progression

Evidence from Turkey

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Motivation

1. Impact of immigration on native fertility is studied, but **not in the context of refugees**.
2. Fertility rates in OECD countries have fallen by half over the past 60 years to roughly **1.5 children per woman**.
3. **Turkey: a unique case study**
 - Host to one of the world's **largest refugee populations** ($\approx 5.3\text{M}$ Syrians registered, $\approx 3.6\text{M}$ in Turkey).
 - Native total fertility now **below the OECD average** (≈ 1.41).
 - Ideal setting for studying how migration shocks interact with **low fertility**.

Research question. How does the influx of Syrian refugees affect the fertility decisions of native Turkish women?

We look at both **marriage** outcomes and **child-parity** outcomes (one to five children) at economically meaningful ages.

Findings.

- **Pooled cells do not survive Romano–Wolf** familywise correction across the 26-cell parity-by-age family.
- **No effect on marriage or early-parity fertility** (parities 1–3, all ages).
- Raw 2SLS: large positive estimates at **higher parities (4–5)** in the **late 20s / 30s**.
- Response **concentrates in three subgroups**: Arabic/Kurdish mother tongue, lower-education, and high-exposure East/Southeast border region.
- **Arabic/Kurdish mother tongue is the one subgroup that survives RW**: fourth-parity at age 28 rejects at the 10% level after the stepdown, with two further cells also rejecting.

Two channels we'll come back to.

- **Labor-market / opportunity-cost** (Aksu, Erzan, Kırdar 2022): refugee inflow reduced *native female* employment but not male employment $\Rightarrow$ lower opportunity cost of an extra child without an income hit.
- **Ideal-fertility / norms**: stated ideal- n shifts upward at the *same upper thresholds* and in the *same subgroups* where realised higher-parity fertility responds.

TDHS: nationally representative microdata on women with rich **fertility histories** and **migration histories** for responding women.

Main data: TDHS **2013 & 2018**; woman $\times$ age **pseudo-panel** from age 12 to survey age; province assigned by **2011 residence** (pre-shock).

Cumulative parity outcomes (absorbing-state indicators):

$$Y_{ija}^{(p)} = \mathbb{1}\{\text{parity}_i \geq p \text{ by age } a\}, \quad p \in \{1, 2, 3, 4, 5\}, \quad a \in \{16, 20, 24, 28, 32, 36\}.$$

Plus the marriage analogue $\mathbb{1}\{\text{ever married by age } a\}$.

Exposure: average refugee-to-native ratio over the woman's age window 12 to a :

$$\bar{R}_{ja} = \frac{1}{a - 11} \sum_{s=12}^a R_{j, t(s,i)}.$$

Interpretation. A 1 pp. higher *average annual* ratio over ages 12– a raises $\Pr(\text{parity} \geq p \text{ by } a)$ by $\hat{\beta}^{(p,a)}$ pp.

[Sample details](#)

[Pseudo-panel](#)

[Data sources](#)

Identification — Quasi-Exogenous Distance Variation

- **Challenge.** Refugee location is endogenous: internal migration responds to local labor-market conditions; by 2018 refugees had spread well beyond border provinces.
- **Strategy.** Follow Aksu, Erzan, Kırdar (2022): instrument $\bar{R}_{ja}$ with a shift-share variable based on geographic distance from Syrian to Turkish provinces.

$$Z_{n,t} = \underbrace{\sum_{s=1}^{13} \frac{(1/d_{s,T}) \pi_s}{\sum_{X \in \{T,L,J,I\}} (1/d_{s,X})}}_{\text{share weight (time-invariant)}} \cdot T_t = B_n \times T_t.$$

π_s : pre-war pop. share of Syrian province s ; $d_{n,s}$: distance Syrian s to Turkish n ; T_t : aggregate Syrian refugee stock across four host countries.

Where the variation comes from.

- B_n is a **time-invariant, geography-based** intensity of refugee exposure for province n : provinces closer to Syrian provinces have higher B_n .
- T_t scales it over time as the aggregate Syrian refugee stock grows post-2011, so the variation that identifies β is $B_n \times (\text{post-shock years})$ — a **continuous-treatment difference-in-differences** design.
- **Exclusion restriction:** conditional on FE and trend controls, B_n is **uncorrelated with pre-shock fertility trends** among Turkish natives.

T_t over time

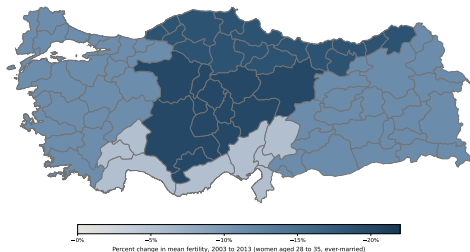
Settlement-Propensity Weight B_n across Provinces

Settlement-propensity weight B_n across Turkish provinces (quintiles)

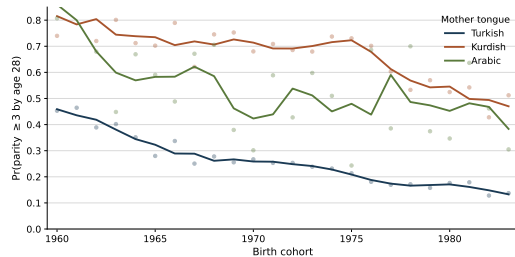


Why Region-Specific and Mother-Tongue-Specific Trends?

Region trends



Mother-tongue trends



For each woman i in province j , we model whether parity p has been reached by age a :

$$Y_{ija}^{(p)} = \alpha + \beta^{(p,a)} \bar{R}_{ja} + X_i' \Gamma + \delta_j + \mu_{t(a,i)} + \theta_{g(j)} \cdot t(a, i) + \lambda_{l(i)} \cdot t(a, i) + \varepsilon_{ija},$$

with $\bar{R}_{ja}$ instrumented by $\bar{Z}_{ja} = B_j \times \bar{T}_a$.

- $\beta^{(p,a)}$ — coefficient of interest: effect of a 1 pp. higher *average annual* refugee-to-native ratio on $\Pr(\text{parity} \geq p \text{ by age } a)$.
- X_i' — individual controls: survey wave, childhood urban/rural, mother's education, mother tongue.
- δ_j — province FE; $\mu_{t(a,i)}$ — year-at-age- a FE ($\equiv$ birth-cohort FE at fixed a).
- $\theta_{g(j)} \cdot t$ — **five-region**-specific linear trends.
- $\lambda_{l(i)} \cdot t$ — **mother-tongue**-specific linear trends (Turkish / Kurdish / Arabic).
- Trends start in 1987; SEs clustered at province; DHS sampling weights.

Placebo — $Y \sim B_n \times t$ on Pre-Shock Cohorts

On the pre-shock subsample (cohort-age window ends before 2012), replace the shift-share instrument $\bar{Z}_{ja}$ with $B_j \times t(a, i)$ — a linear placebo interaction in calendar time. Under the null of no pre-trends in B_j , $\beta(p, a)$ should be zero.

$$Y_{ija}^{(p)} = \alpha + \gamma^{(p,a)}(B_j \times t(a, i)) + X_i' \Gamma + \delta_j + \mu_{t(a,i)} + \theta_{g(j)} t + \varepsilon_{ija}.$$

Outcome	Age 16	Age 20	Age 24	Age 28	Age 32	Age 36
Ever Married	0.0003 (0.0005) [0.84]	0.0010 (0.0007) [0.47]	0.0012* (0.0007) [0.47]	0.0008** (0.0004) [0.38]	0.0011* (0.0006) [0.47]	-0.0001 (0.0007) [0.90]
Ever Gave Birth to One Child	0.0005* (0.0003) [0.77]	0.0011 (0.0008) [0.92]	0.0007 (0.0010) [1.00]	0.0011** (0.0005) [0.62]	0.0011** (0.0005) [0.74]	0.0001 (0.0007) [1.00]
Ever Gave Birth to Two Children	0.0001 (0.0002) [1.00]	-0.0001 (0.0004) [1.00]	0.0009 (0.0006) [0.91]	0.0018*** (0.0007) [0.48]	0.0015 (0.0010) [0.91]	0.0006 (0.0011) [1.00]
Ever Gave Birth to Three Children	0.0000 (0.0000) [0.92]	-0.0000 (0.0002) [1.00]	0.0004 (0.0007) [1.00]	0.0000 (0.0008) [1.00]	-0.0007 (0.0008) [0.99]	-0.0024* (0.0013) [0.79]
Ever Gave Birth to Four Children	— (—)	— (—)	0.0002 (0.0004) [1.00]	0.0011 (0.0009) [0.92]	0.0005 (0.0023) [1.00]	-0.0005 (0.0022) [1.00]
Ever Gave Birth to Five Children	— (—)	— (—)	0.0000 (0.0002) [1.00]	0.0002 (0.0005) [1.00]	0.0012 (0.0014) [0.99]	0.0018 (0.0022) [0.99]
Observations	12,355	11,560	10,226	8,194	6,063	4,098

Near-zero $\hat{\gamma}^{(p,a)}$. Stars = raw inference; **[brackets]** = Romano–Wolf stepdown p -values across the 32-cell family. Under **Romano–Wolf**, no cell rejects.

What is RW?

First stage

NUTS-1 placebo

NUTS-2 placebo

Main Results — Primary Sample, Romano–Wolf p -values

Outcome	Age 16	Age 20	Age 24	Age 28	Age 32	Age 36
Ever Married	-0.047 (0.217) [1.00]	-0.572 (0.600) [0.90]	0.513 (1.689) [1.00]	-0.013 (1.047) [1.00]	0.220 (0.920) [1.00]	-0.299 (1.241) [1.00]
Ever Birth ≥ 1	0.010 (0.127) [1.00]	0.123 (0.523) [1.00]	-0.043 (1.968) [1.00]	0.923 (1.342) [1.00]	-1.515 (1.789) [0.99]	0.184 (1.371) [1.00]
Ever Birth ≥ 2	0.028 (0.041) [1.00]	0.305 (0.546) [1.00]	-0.151 (0.746) [1.00]	0.613 (1.906) [1.00]	1.740 (3.419) [1.00]	-0.845 (1.189) [1.00]
Ever Birth ≥ 3	0.008 (0.005) [0.88]	0.382 (0.248) [0.88]	-0.371 (0.565) [1.00]	1.250 (1.559) [1.00]	2.893 (2.700) [0.98]	1.062 (1.810) [1.00]
Ever Birth ≥ 4	– (–)	– (–)	0.250 (0.340) [1.00]	1.905** (0.809) [0.52]	2.246 (1.415) [0.86]	4.252** (1.963) [0.61]
Ever Birth ≥ 5	– (–)	– (–)	0.313 (0.232) [0.92]	1.392* (0.803) [0.81]	1.414 (0.912) [0.88]	4.133** (1.856) [0.58]
Observations	14,354	13,484	12,241	10,240	8,245	6,110

Stars = raw inference; **[brackets]** = Romano–Wolf stepdown p -values across the 32-cell family. Positive raw effects at parities 4–5, ages 28+. **No cell survives RW** in the pooled sample.

What is RW?

4-wave EM

OLS

Robustness

Heterogeneity — Lower-Educated Women (RW p -values)

Outcome	Age 16	Age 20	Age 24	Age 28	Age 32	Age 36
<i>Lower education (no schooling / primary)</i>						
Ever Married	-0.185 (0.303) [0.95]	-0.872 (1.285) [0.95]	2.048 (1.824) [0.86]	-0.059 (1.529) [1.00]	0.189 (1.035) [1.00]	-0.203 (1.190) [1.00]
Ever Birth ≥ 1	-0.023 (0.187) [1.00]	0.048 (1.249) [1.00]	0.291 (2.443) [1.00]	1.303 (1.957) [1.00]	0.045 (1.274) [1.00]	0.558 (1.331) [1.00]
Ever Birth ≥ 2	0.003 (0.057) [1.00]	0.261 (1.636) [1.00]	-1.116 (1.682) [1.00]	1.691 (1.645) [0.98]	1.602 (2.743) [1.00]	1.986 (1.635) [0.96]
Ever Birth ≥ 3	0.004 (0.007) [1.00]	0.738 (0.765) [0.99]	-0.637 (1.210) [1.00]	2.823 (2.238) [0.96]	3.261 (2.446) [0.94]	4.689*** (1.697) [0.33]
Ever Birth ≥ 4	— (—)	— (—)	-0.067 (0.679) [1.00]	3.175** (1.294) [0.45]	2.968*** (1.053) [0.32]	6.658*** (2.370) [0.32]
Ever Birth ≥ 5	— (—)	— (—)	0.351 (0.414) [0.99]	1.787 (1.284) [0.93]	1.893 (1.160) [0.85]	5.619*** (1.860) [0.26]
Observations	9,361	9,132	8,729	7,633	6,348	4,897

Those with primary school or less. Sharp positive raw effects at parities **4–5, ages 28–36**; bracketed RW p -values **rise substantially** (smallest 0.26); **no cell rejects at $\alpha = 0.05$** .

Baseline rates

EM analog

Higher-educated

Heterogeneity — Kurdish / Arabic Mother Tongue (RW p -values)

Outcome	Age 16	Age 20	Age 24	Age 28	Age 32	Age 36
<i>Arabic / Kurdish mother tongue</i>						
Ever Married	0.556* (0.323) [0.53]	0.520 (1.759) [0.96]	0.585 (2.478) [0.96]	-4.025* (2.239) [0.52]	1.933 (2.238) [0.87]	0.859 (1.146) [0.87]
Ever Birth ≥ 1	0.336* (0.177) [0.73]	1.821 (1.593) [0.95]	-1.754 (2.323) [0.99]	-1.585 (2.396) [0.99]	2.441 (2.770) [0.98]	0.762 (1.811) [0.99]
Ever Birth ≥ 2	0.047 (0.073) [0.99]	1.806 (1.166) [0.82]	1.433 (1.838) [0.99]	1.460 (2.987) [0.99]	4.418 (3.288) [0.89]	1.994 (1.401) [0.86]
Ever Birth ≥ 3	-0.019 (0.026) [0.99]	0.647 (1.229) [0.99]	-0.869 (1.230) [0.99]	12.305*** (2.469) [0.04]	6.019* (3.479) [0.78]	2.569* (1.522) [0.79]
Ever Birth ≥ 4	— (—)	— (—)	1.201 (1.318) [0.98]	8.119*** (1.752) [0.05]	2.024 (3.811) [0.99]	8.244*** (1.788) [0.05]
Ever Birth ≥ 5	— (—)	— (—)	0.394 (1.017) [0.99]	3.155 (2.450) [0.92]	4.719 (3.347) [0.86]	5.038** (2.366) [0.64]
Observations	3,145	2,785	2,415	1,951	1,489	1,055

One subgroup that survives Romano–Wolf: parity 3@28 ($p^{\text{RW}} = 0.04$), parity 4@28 ($p^{\text{RW}} = 0.05$), parity 4@36 ($p^{\text{RW}} = 0.05$). Turkish-speaking women null.

Baseline rates

EM analog

Turkish MT

Heterogeneity — High-Exposure Border Region (RW p -values)

Outcome	Age 16	Age 20	Age 24	Age 28	Age 32	Age 36
<i>High-exposure border region (East / Southeast)</i>						
Ever Married	-0.245 (0.358) [0.87]	-1.691*** (0.646) [0.27]	-2.591 (1.976) [0.67]	-0.913 (1.656) [0.87]	-0.430 (0.761) [0.87]	1.565* (0.855) [0.50]
Ever Birth ≥ 1	-0.023 (0.154) [1.00]	-0.404 (0.478) [0.98]	-3.030* (1.569) [0.65]	1.446 (2.261) [0.99]	0.339 (1.428) [1.00]	1.989* (1.185) [0.70]
Ever Birth ≥ 2	0.021 (0.052) [1.00]	1.425*** (0.471) [0.32]	1.283 (0.934) [0.83]	0.470 (2.126) [1.00]	5.495** (2.469) [0.56]	0.330 (1.105) [1.00]
Ever Birth ≥ 3	-0.015 (0.019) [0.98]	0.339 (0.429) [0.98]	-0.050 (0.599) [1.00]	5.112** (2.045) [0.49]	9.121*** (2.816) [0.29]	4.400** (1.763) [0.49]
Ever Birth ≥ 4	— (—)	— (—)	0.229 (0.575) [1.00]	2.987** (1.283) [0.55]	6.751*** (1.698) [0.15]	5.753** (2.491) [0.55]
Ever Birth ≥ 5	— (—)	— (—)	-0.014 (0.427) [1.00]	1.875** (0.934) [0.62]	4.148*** (1.390) [0.33]	6.594* (3.499) [0.65]
Observations	4,002	3,522	3,022	2,404	1,851	1,311

East / Southeast (highest B_n , ~55% Kurdish-speaking). Largest raw point estimates at parities **3–5**, **ages 24+**; RW p -values rise but cells do not reject at $\alpha = 0.05$. Rest-of-Turkey null.

Baseline rates

EM analog

Rest of Turkey

Mechanism I — Labor-Market / Opportunity-Cost Channel

For $Y_i \in \{W_i, H_i\}$ (woman currently employed; husband worked last 7 days), on the TDHS women whose fertility we study, with $\bar{R}$ instrumented by $B_{j(i)} \bar{T}_{a_i^*}$:

$$Y_i = \beta \bar{R}_{j(i), a_i^*} + \alpha_{j(i)} + \mu_{t_i} + X_i' \gamma + \sum_r \lambda_r D_{it}^r + \varepsilon_i.$$

Subgroup	(1) Woman employed	(2) Husband worked 7d
Pooled	-2.7**	+2.0*
Lower education	-4.8***	+2.8**
Arabic / Kurdish mother tongue	-3.9**	≈ 0
High-exposure border region (E/SE)	-2.2*	≈ 0

Gender-asymmetric pattern: female employment falls in the pooled sample and in every subgroup where the higher-parity fertility response is concentrated; husband employment is stable or higher — same signature as Aksu, Erzan, Kırdar (2022).

Mechanism II — Ideal Number of Children, by Subgroup

Cross-sectional 2SLS of stated ideal n_i^* (and threshold cuts $\mathbb{1}\{n_i^* \geq k\}$) on $\bar{R}_{j(i),t(i)}$, ever-married four-wave sample. Same controls / FE / trends as the fertility specification.

Subgroup	Primary	Threshold decomposition			
	Ideal n	Ideal ≥ 2	Ideal ≥ 3	Ideal ≥ 4	Ideal ≥ 5
Full sample (pooled)	1.197** (0.541)	0.046 (0.061)	0.174 (0.254)	0.514* (0.270)	0.256** (0.109)
Lower education	1.533*** (0.549)	0.001 (0.088)	0.203 (0.247)	0.676*** (0.247)	0.351*** (0.101)
Arabic / Kurdish mother tongue	3.034** (1.253)	0.129 (0.158)	0.512 (0.417)	0.911*** (0.350)	0.475*** (0.184)
High-exposure border region	1.566** (0.734)	0.069 (0.117)	0.212 (0.237)	0.727** (0.336)	0.368** (0.143)

Stated ideal n shifts up at the upper thresholds (≥ 4 , ≥ 5), strongest in the three treated subgroups (low-educ, Arabic/Kurdish, high-exposure border region). A shift in *desired* family size is hard to read as pure labor-market response: complementary **norms** / **preference** channel.

The Syrian inflow raised native parity progression in Kurdish/Arabic, lower-educated, and southeast subgroups.

- **No effect** on marriage or first births anywhere.
- Pooled higher-parity (4–5) estimates are large and positive under raw inference but **do not survive Romano–Wolf** across 32 tests.
- Effects **concentrate** in three overlapping groups: **Kurdish/Arabic mother tongue, lower-educated, heavily-treated southeast**. Arabic/Kurdish subgroup survives RW.
- Pattern (parity-progression, not entry into motherhood) is most consistent with a **gender-asymmetric labor-market** channel; ideal- n shifts at upper thresholds add a complementary norms reading.

Thank you. / Questions?

Appendix

Appendix: Romano–Wolf Multiple-Testing Correction

The problem. The main fertility table is a family of **32 hypotheses** (5 parities $\times$ 6 ages + 2 marriage cells). At nominal $\alpha = 0.05$, ~ 1.6 false rejections are expected by chance even if no real effect exists.

What RW does. Adjusts each p -value so that the probability of *any* false rejection across the whole family stays below α – the **familywise error rate** (FWER):

$$\text{FWER} = \Pr(\text{at least one false rejection}) \leq \alpha.$$

How. Bootstrap the joint distribution of t -statistics – preserving province-level clustering and cross-cell correlation – and adjust each cell's p -value to the bootstrap probability that the most extreme remaining t -stat exceeds the observed one (**stepdown**).

What changes in the tables. Cells with raw $p < 0.05$ may have RW $p > 0.05$ – the bracketed values rise; this is the price of testing 32 hypotheses simultaneously.

Appendix ● Four-wave EM sample

Main + three subgroups, TDHS 2003/08/13/18, childhood province.

Appendix: Main IV — Four-Wave Ever-Married Sample (RW)

Main IV (2SLS) results, ever-married sample.

Outcome	Age 16	Age 20	Age 24	Age 28	Age 32	Age 36
Ever Gave Birth to One Child	0.313 (1.914) [1.00]	-1.348 (1.229) [0.98]	0.986 (2.238) [1.00]	-0.608 (1.109) [1.00]	-1.397 (1.437) [0.99]	1.373** (0.680) [0.74]
Ever Gave Birth to Two Children	0.322* (0.172) [0.81]	1.273 (2.240) [1.00]	0.687 (1.452) [1.00]	-1.075 (2.018) [1.00]	2.253 (2.615) [0.99]	0.100 (1.927) [1.00]
Ever Gave Birth to Three Children	0.143 (0.117) [0.96]	1.482 (0.974) [0.91]	-0.772 (0.924) [0.99]	-0.900 (1.991) [1.00]	0.714 (2.365) [1.00]	-0.561 (2.507) [1.00]
Ever Gave Birth to Four Children	— (—)	— (—)	-0.368 (0.518) [1.00]	0.401 (0.754) [1.00]	0.029 (1.582) [1.00]	2.742 (2.004) [0.94]
Ever Gave Birth to Five Children	— (—)	— (—)	0.280 (0.305) [0.99]	0.789 (0.551) [0.92]	-0.346 (1.274) [1.00]	1.567 (1.624) [0.99]
Observations	19,281	21,525	21,782	19,776	16,442	12,282

Ever-married, TDHS 2003/08/13/18, childhood province. Trade-off: more pre-shock anchoring but (i) marriage is treatment-responsive (selection); (ii) measurement error on province.

Appendix: Lower-Educated Women — Four-Wave EM Sample

Main IV (2SLS) results, ever-married sample.

Outcome	Age 16	Age 20	Age 24	Age 28	Age 32	Age 36
<i>Lower-educated women ($\leq$ primary)</i>						
Ever Birth ≥ 1	-0.452 (2.179) [1.00]	-1.843 (1.977) [0.99]	-0.507 (1.713) [1.00]	0.527 (1.357) [1.00]	0.251 (0.958) [1.00]	1.277 (0.888) [0.91]
Ever Birth ≥ 2	0.145 (0.237) [1.00]	0.151 (2.977) [1.00]	-1.999 (2.432) [1.00]	0.419 (1.279) [1.00]	1.410 (2.445) [1.00]	1.543 (1.342) [0.97]
Ever Birth ≥ 3	0.002 (0.154) [1.00]	1.480 (1.370) [0.97]	-1.751 (1.414) [0.97]	0.858 (2.529) [1.00]	1.391 (1.985) [1.00]	1.931 (2.720) [1.00]
Ever Birth ≥ 4	— (—)	— (—)	-0.605 (0.858) [1.00]	1.699 (1.219) [0.95]	1.063 (1.205) [0.99]	4.051 (2.468) [0.84]
Ever Birth ≥ 5	— (—)	— (—)	0.350 (0.412) [0.99]	1.148 (0.882) [0.96]	0.151 (1.468) [1.00]	2.733* (1.749) [0.84]
Observations	14,711	16,562	16,952	15,800	13,465	10,305

Positive higher-parity raw estimates at ages 28+ persist; no cell survives RW in the EM family.

Appendix: Kurdish / Arabic Mother Tongue — Four-Wave EM Sample

Main IV (2SLS) results, ever-married sample.

Outcome	Age 16	Age 20	Age 24	Age 28	Age 32	Age 36
<i>Arabic / Kurdish mother tongue</i>						
Ever Birth ≥ 1	1.473 (1.219) [0.92]	-1.001 (2.029) [1.00]	-1.113 (2.688) [1.00]	-2.204 (2.129) [0.95]	1.714 (1.892) [0.96]	0.322 (1.488) [1.00]
Ever Birth ≥ 2	0.251 (0.412) [1.00]	2.834* (1.771) [0.55]	2.140 (3.603) [1.00]	0.810 (2.459) [1.00]	2.998 (2.431) [0.90]	2.097* (1.582) [0.90]
Ever Birth ≥ 3	-0.082 (0.329) [1.00]	2.023 (1.993) [0.92]	-1.391 (1.874) [1.00]	12.349*** (2.473) [0.05]	3.253 (3.036) [0.96]	1.314 (2.216) [1.00]
Ever Birth ≥ 4	— (—)	— (—)	-0.184 (1.792) [1.00]	8.438** (2.470) [0.43]	2.087 (3.854) [1.00]	7.109*** (1.750) [0.17]
Ever Birth ≥ 5	— (—)	— (—)	0.306 (0.985) [1.00]	4.181 (2.658) [0.90]	6.160* (2.870) [0.75]	6.020** (2.091) [0.32]
Observations	4,492	4,765	4,517	3,885	3,041	2,157

The parity $\geq 3@28$ cell rejects at $\alpha = 0.05$ even in the EM sample ($p^{\text{RW}} \approx 0.05$). Higher-parity raw estimates at 4@28, 4@36, 5@28 are large but do not survive RW in the EM family.

Appendix: High-Exposure Border Region — Four-Wave EM Sample

Main IV (2SLS) results, ever-married sample.

Outcome	Age 16	Age 20	Age 24	Age 28	Age 32	Age 36
<i>High-exposure border region (E/SE)</i>						
Ever Birth ≥ 1	3.464** (1.693) [0.36]	-0.142 (1.035) [1.00]	-0.700 (0.997) [0.97]	1.105 (1.828) [0.96]	1.292 (1.170) [0.85]	0.781 (0.982) [0.92]
Ever Birth ≥ 2	0.723 (0.312) [0.72]	4.920*** (2.204) [0.17]	3.907*** (2.012) [0.39]	-0.081 (1.860) [1.00]	6.482*** (2.197) [0.17]	0.783 (1.061) [0.96]
Ever Birth ≥ 3	0.226 (0.182) [0.89]	1.177 (1.303) [0.88]	-0.264 (1.378) [1.00]	4.832* (2.055) [0.54]	9.565*** (2.461) [0.06]	4.237* (1.606) [0.60]
Ever Birth ≥ 4	— (—)	— (—)	0.568 (1.068) [0.97]	3.187 (1.430) [0.66]	7.355*** (1.890) [0.10]	8.343 (2.308) [0.60]
Ever Birth ≥ 5	— (—)	— (—)	0.331 (0.587) [0.97]	2.436** (1.102) [0.54]	4.365 (1.601) [0.64]	7.321** (3.818) [0.31]
Observations	5,666	6,103	5,823	4,995	3,954	2,796

Raw higher-parity point estimates at ages 28+ remain large and positive, but **no EM cell survives RW at $\alpha = 0.05$** .

Appendix ● Baseline parity-attainment by subgroup

Share of women in each age column who have reached the row outcome.

Appendix: Baseline Parity-Attainment — Lower-Educated Women

Outcome	Age 16	Age 20	Age 24	Age 28	Age 32	Age 36
Ever Married	16.1%	61.5%	85.5%	93.5%	96.0%	97.4%
Ever Gave Birth to One Child	5.4%	44.7%	76.1%	87.9%	92.6%	94.5%
Ever Gave Birth to Two Children	0.7%	15.4%	47.1%	69.9%	81.4%	86.5%
Ever Gave Birth to Three Children	0.0%	2.9%	16.0%	32.5%	46.7%	53.6%
Ever Gave Birth to Four Children	0.0%	0.7%	5.5%	14.2%	21.2%	26.6%
Ever Gave Birth to Five Children	0.0%	0.1%	1.6%	6.3%	10.8%	14.4%
Observations	8,316	8,514	8,468	7,739	6,477	5,026

Higher baseline higher-parity rates than the overall sample (e.g. parity ≥ 4 by 32: 21.2% vs. 15.9% overall) — room to move up the parity ladder.

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Appendix: Baseline Parity-Attainment — Kurdish / Arabic Mother Tongue

Outcome	Age 16	Age 20	Age 24	Age 28	Age 32	Age 36
Ever Married	16.6%	55.0%	78.5%	89.6%	93.6%	96.4%
Ever Gave Birth to One Child	5.5%	41.3%	69.7%	84.0%	90.5%	93.6%
Ever Gave Birth to Two Children	1.2%	20.4%	54.0%	74.1%	84.6%	90.9%
Ever Gave Birth to Three Children	0.1%	6.0%	28.8%	53.6%	69.4%	78.7%
Ever Gave Birth to Four Children	0.0%	1.8%	14.1%	33.9%	48.3%	60.6%
Ever Gave Birth to Five Children	0.0%	0.3%	4.7%	18.9%	31.8%	43.8%
Observations	2,940	2,722	2,396	1,987	1,526	1,090

Substantially higher baseline higher-parity rates (e.g. parity ≥ 4 by 32: 44.2% vs. 15.9% overall; parity ≥ 5 by 32: 28.7% vs. 8.0%). Largest scope for upward parity progression.

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Appendix: Baseline Parity-Attainment — High-Exposure Border Region

Outcome	Age 16	Age 20	Age 24	Age 28	Age 32	Age 36
Ever Married	12.9%	50.6%	75.3%	87.7%	92.9%	95.4%
Ever Gave Birth to One Child	4.6%	36.5%	65.5%	82.0%	89.6%	92.4%
Ever Gave Birth to Two Children	0.9%	17.3%	48.1%	69.6%	81.6%	88.2%
Ever Gave Birth to Three Children	0.1%	4.8%	24.5%	48.8%	64.8%	74.9%
Ever Gave Birth to Four Children	0.0%	1.4%	11.9%	29.7%	44.2%	58.2%
Ever Gave Birth to Five Children	0.0%	0.3%	3.8%	15.9%	28.7%	41.0%
Observations	3,757	3,406	2,981	2,453	1,902	1,350

Highest baseline higher-parity rates (e.g. parity ≥ 5 by 32: 28.7% – shared with the Kurdish/Arabic subgroup, since the two cuts overlap heavily in the East/Southeast). Common thread across all three heterogeneity slides: **effects appear where the parity ladder has room left to climb.**

Appendix • Robustness

One slide per check; each returns to slide 10 (Main Results).

NUTS-1 trends

NUTS-2 trends

Drop post-2011 movers

4-wave decomposition

OLS counterpart

Event study (pre-trends)

Parity-progression hazards

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Appendix: Robustness — NUTS-1-Region Linear Trends

Outcome	Age 16	Age 20	Age 24	Age 28	Age 32	Age 36
Ever Married	-0.122 (0.247) [0.98]	-0.510 (0.683) [0.95]	0.414 (1.832) [0.98]	0.230 (1.070) [0.98]	0.859 (0.980) [0.93]	-0.413 (1.274) [0.98]
Ever Gave Birth to One Child	-0.020 (0.147) [1.00]	0.305 (0.587) [1.00]	-0.027 (2.139) [1.00]	1.051 (1.342) [1.00]	-1.000 (1.828) [1.00]	0.381 (1.528) [1.00]
Ever Gave Birth to Two Children	0.029 (0.050) [1.00]	0.359 (0.591) [1.00]	-0.126 (0.806) [1.00]	0.628 (2.184) [1.00]	1.726 (3.593) [1.00]	-0.883 (1.220) [1.00]
Ever Gave Birth to Three Children	0.004 (0.006) [1.00]	0.446** (0.224) [0.73]	-0.382 (0.595) [1.00]	0.769 (1.670) [1.00]	2.722 (2.679) [0.98]	1.327 (1.767) [1.00]
Ever Gave Birth to Four Children	— (—)	— (—)	0.557 (0.403) [0.93]	1.731** (0.849) [0.72]	1.647 (1.471) [0.97]	4.184* (2.159) [0.73]
Ever Gave Birth to Five Children	— (—)	— (—)	0.304 (0.241) [0.95]	1.601* (0.919) [0.82]	1.108 (0.911) [0.95]	4.020** (1.965) [0.72]
Observations	14,354	13,484	12,241	10,240	8,245	6,110

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Appendix: Robustness — NUTS-2-Region Linear Trends

Outcome	Age 16	Age 20	Age 24	Age 28	Age 32	Age 36
Ever Married	-0.288 (0.289) [0.84]	-1.093 (0.802) [0.68]	-0.758 (1.920) [0.90]	-1.000 (1.010) [0.84]	-0.129 (1.139) [0.90]	-0.757 (1.184) [0.90]
Ever Gave Birth to One Child	-0.107 (0.182) [1.00]	-0.264 (0.771) [1.00]	-1.124 (2.515) [1.00]	0.022 (1.436) [1.00]	-2.168 (1.822) [0.95]	-0.729 (1.575) [1.00]
Ever Gave Birth to Two Children	0.017 (0.039) [1.00]	-0.100 (0.457) [1.00]	-0.811 (1.015) [0.99]	-1.065 (2.557) [1.00]	-0.480 (3.478) [1.00]	-2.465 (1.582) [0.87]
Ever Gave Birth to Three Children	0.000 (0.006) [1.00]	0.298 (0.214) [0.90]	-0.755 (0.761) [0.98]	0.744 (1.759) [1.00]	1.413 (2.806) [1.00]	1.705 (1.959) [0.99]
Ever Gave Birth to Four Children	— (—)	— (—)	0.243 (0.411) [1.00]	1.630** (0.831) [0.74]	0.778 (2.428) [1.00]	3.357 (2.636) [0.93]
Ever Gave Birth to Five Children	— (—)	— (—)	0.232 (0.227) [0.98]	1.295 (0.871) [0.89]	1.345 (1.194) [0.96]	3.523 (3.186) [0.96]
Observations	14,354	13,484	12,241	10,240	8,245	6,110

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Appendix: Robustness — Dropping Post-2011 Out-Migrants

	Parity ≥ 4		Parity ≥ 5	
	by 28	by 32	by 28	by 32
Primary sample	1.905** (0.809)	2.246 (1.415)	1.392* (0.803)	1.414 (0.912)
<i>N</i>	10,240	8,245	10,240	8,245
Drop post-2011 movers	3.220*** (1.101)	3.914* (2.059)	1.668 (1.177)	1.399 (1.026)
<i>N</i>	9,365	7,643	9,365	7,643

Concern: women in high- B_n provinces in 2011 may have moved away, attenuating exposure for those who stayed. Movers = 12.3% pooled (7.25% in 2013, 19.53% in 2018). Mover indicator **uncorrelated with** B_n ($\hat{\beta} = 0.002$, $p = 0.73$). Non-mover estimates are **larger and more precise** at the four core higher-parity cells, consistent with cleaner exposure measurement among stayers.

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Appendix: Robustness — Four-Wave Decomposition Ladder

Core outcome	(i) Primary 2013/18, full, prov. 2011	(ii) EM, prov. 2011	(iii) EM, childhood prov.	(iv) Four-wave 2003–18, EM, childhood prov.
Ever Gave Birth to Four Children, age 28	1.905** (0.809)	1.768** (0.879)	1.750* (0.940)	1.211 (1.007)
Ever Gave Birth to Four Children, age 32	2.246 (1.415)	2.260 (1.375)	2.098 (1.543)	1.198 (1.517)
Ever Gave Birth to Five Children, age 28	1.392* (0.803)	1.313 (0.851)	0.747 (0.749)	1.228 (0.997)
Ever Gave Birth to Five Children, age 32	1.414 (0.912)	1.437 (0.924)	1.159 (1.238)	1.099 (1.067)
Observations (egb4, age 28)	10,240	9,746	9,746	19,776

Decomposes the gap between primary FS estimates and four-wave EM estimates by changing one feature at a time.

- (i)→(ii): ever-married restriction (small effect).
- (ii)→(iii): replacing 2011 province with childhood province (**attenuates parity 5028 from 1.31 to 0.75** — measurement error).
- (iii)→(iv): adds 2003/08 waves (further attenuates; less precision).

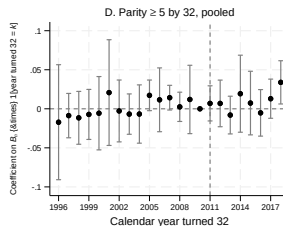
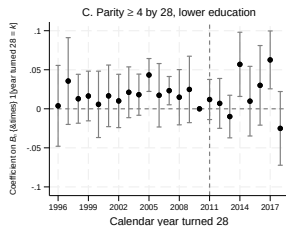
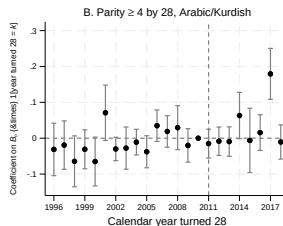
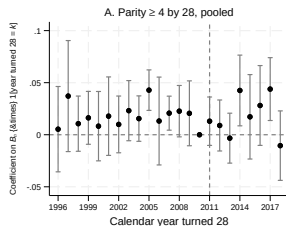
Reading: **FS spec is preferred baseline**; four-wave EM is a complementary lower-bound check.

Appendix: Robustness — OLS Counterpart of the Main 2SLS

Outcome	Age 16	Age 20	Age 24	Age 28	Age 32	Age 36
Ever Married	-0.036 (0.176)	-0.743 (0.635)	0.799 (1.472)	-1.056 (1.458)	-0.414 (0.758)	-0.164 (0.862)
Ever Gave Birth to One Child	-0.013 (0.131)	-0.259 (0.418)	0.655 (1.327)	-0.185 (1.561)	-1.190 (1.145)	-0.478 (0.975)
Ever Gave Birth to Two Children	0.036 (0.029)	0.443 (0.391)	0.606 (0.788)	0.320 (1.584)	0.080 (2.349)	-0.220 (1.091)
Ever Gave Birth to Three Children	0.006* (0.003)	0.128 (0.319)	-0.516 (0.429)	1.592 (1.387)	2.069 (2.045)	2.311* (1.369)
Ever Gave Birth to Four Children	— (—)	— (—)	0.050 (0.263)	1.367 (0.909)	2.565** (1.130)	3.543** (1.543)
Ever Gave Birth to Five Children	— (—)	— (—)	0.020 (0.353)	0.808 (0.718)	2.888*** (0.742)	1.359 (1.301)
Observations	14,354	13,484	12,241	10,240	8,245	6,110

OLS estimates on the same sample / controls / trends as the main 2SLS. **Coefficients are close in magnitude to the 2SLS**: the four core higher-parity cells differ by less than ~ 1.5 between OLS and 2SLS, and the sign / pattern is unchanged. The IV machinery is correcting selection-on-province bias but is **not generating the headline pattern**; the same higher-parity signature appears in OLS.

Appendix: Robustness — Event Study around 2011



Appendix: Robustness — Parity-Progression Hazard Decomposition

Discrete-time hazard 2SLS on at-risk person-years for marriage and each birth transition $k \in \{1, \dots, 5\}$; $R_{j(i,t),t}$ instrumented by $Z_{j(i,t),t} = B_{j(i,t)} \times T_t$:

$$h_{ijt}^{(k)} = \alpha^{(k)} + \beta^{(k)} R_{j(i,t),t} + X_i' \Gamma^{(k)} + \phi_{\text{age}(i,t)}^{(k)} + D_k \psi_{\tau(i,t)}^{(k)} + \delta_{j(i,t)}^{(k)} + \mu_t^{(k)} + \theta_{g(j)}^{(k)} t + \varepsilon_{ijt}^{(k)}.$$

	Marriage	1st Birth	2nd Birth	3rd Birth	4th Birth	5th Birth
Refugee ratio R_{jt}	0.0396 (0.0587) [0.94]	-0.0568 (0.0426) [0.67]	-0.0324 (0.1880) [1.00]	0.0638 (0.0775) [0.90]	-0.0056 (0.0677) [1.00]	0.0100 (0.0616) [1.00]
Person-years	142,371	166,710	54,655	63,252	33,962	15,459

Marriage and first-birth hazards essentially zero; higher-parity hazards small and noisy. Current-year exposure on person-years cannot localize the cumulative response transition by transition. **Consistent with, not independently confirming**, the cumulative higher-parity story.

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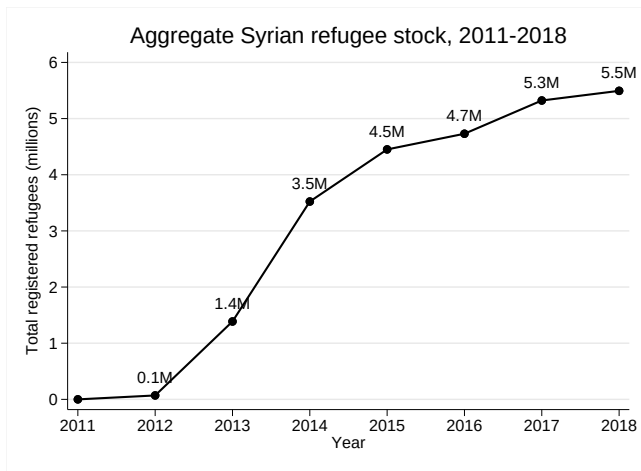
Appendix ● Setting, sample, data

Appendix: First-Stage F and Coefficient on $\bar{Z}_{ja}$

	Age 16	Age 20	Age 24	Age 28	Age 32	Age 36
<i>Panel A: Primary sample (2013/18, full sample, province in 2011)</i>						
$\bar{Z}_{ja} (\times 10^{-5})$	2.514*** (0.379)	2.229*** (0.195)	2.269*** (0.224)	2.453*** (0.371)	2.293*** (0.315)	2.310*** (0.193)
First-stage F	43.9	131.3	102.4	43.7	52.9	143.7
Observations	14,354	13,484	12,241	10,240	8,245	6,110
<i>Panel B: Four-wave sample (2003/08/13/18, ever-married, childhood province)</i>						
$\bar{Z}_{ja} (\times 10^{-5})$	2.222*** (0.410)	2.359*** (0.411)	2.389*** (0.334)	2.430*** (0.368)	2.251*** (0.309)	2.338*** (0.256)
First-stage F	29.4	32.9	51.2	43.5	53.0	83.2
Observations	19,281	21,525	21,782	19,776	16,442	12,282

First-stage $F \in [43, 144]$ across all primary-sample cells; $F \in [29, 83]$ in the four-wave EM sample. Instrument is **strong** across the board; weak-IV concerns do not bind.

Appendix: Aggregate Refugee Stock T_t across Host Countries



T_t builds up progressively after 2011 across Turkey, Lebanon, Jordan, and Iraq, rather than as a single step change. The shock activates B_n gradually over the 2011–2018 window.

Primary sample (full): TDHS 2013 + 2018.

- Ever- and never-married women aged 15–49 at interview.
- Province assigned by **2011 residence** (pre-shock, predetermined).
- Full controls available (wave, childhood urban/rural, mother's education, mother tongue).

Four-wave EM (appendix): TDHS 2003 + 2008 + 2013 + 2018.

- Ever-married only (1998/2003/2008 surveyed only ever-married women).
- Province by **childhood province** (no detailed adult migration history in 2003/08).
- Used as complementary specification; subject to selection + measurement error.

Why FS leads. Marriage is itself a treatment-responsive outcome (selection); classical measurement error on province $\Rightarrow$ attenuation bias toward zero.

Appendix: Reconstructing the Pseudo-Panel

Cross-sectional record (TDHS interview):

ID	Birth Yr	Childhood Prov	Move Yr	Dest. Prov	Birth 1	Birth 2
10001	1990	Adana	2009	Mardin	2010	2013

⇓ expanded into woman-by-year panel ⇓

Pseudo-panel:

ID	Age	Year	Province	Gave Birth 1	Gave Birth 2
10001	12	2002	Adana	0	0
⋮	⋮	⋮	⋮	⋮	⋮
10001	20	2010	Mardin	1	0
10001	23	2013	Istanbul	1	1

Outcomes are collapsed to one row per (i, a) for each age cutoff; exposure $\bar{R}_{ja}$ averaged over age 12– a .

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Province-year ratio. $R_{j,t}$ = registered Syrians / native population in province j , year t , from **Ministry of Interior** / **AFAD** administrative records (2012–2018; zero before 2012).

Cohort variation in $\bar{R}_{ja}$. A 2013 woman observed at age 28 averages exposure over 1997–2013 (**2 post-shock years**); a 2018 woman at age 28 averages over 2002–2018 (**7 post-shock years**). Same age, same province, *different* $\bar{R}_{ja}$. Crossed with B_j , this cross-cohort variation in window timing is what the IV uses on top of FE.

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Appendix ● Complement subgroups

Panel B of paper heterogeneity tables: the other half of each cut.

Appendix: Higher-Educated Women

Outcome	Age 16	Age 20	Age 24	Age 28	Age 32	Age 36
<i>Higher education (secondary / tertiary)</i>						
Ever Married	0.018 (0.053) [0.99]	0.214 (0.589) [0.99]	-0.886 (3.014) [0.99]	-1.412 (3.269) [0.99]	-1.633 (2.920) [0.99]	1.927 (4.802) [0.99]
Ever Birth ≥ 1	0.041 (0.029) [0.94]	-0.391 (0.672) [0.99]	-0.579 (2.911) [1.00]	0.481 (3.708) [1.00]	-9.425* (4.929) [0.85]	0.625 (5.498) [1.00]
Ever Birth ≥ 2	0.001 (0.024) [1.00]	0.304 (0.239) [0.96]	0.274 (1.155) [1.00]	0.100 (4.655) [1.00]	-4.555 (7.553) [0.99]	-15.097* (8.562) [0.88]
Ever Birth ≥ 3	- (-) [0.96]	0.080 (0.061) [0.96]	0.044 (0.806) [1.00]	-3.229** (1.266) [0.67]	7.384 (6.983) [0.97]	-18.786** (7.432) [0.67]
Ever Birth ≥ 4	- (-) [0.97]	- (-) [0.97]	-0.571 (0.497) [0.97]	-0.322 (0.370) [0.98]	3.927 (3.189) [0.96]	-8.802* (4.863) [0.87]
Ever Birth ≥ 5	- (-) [0.97]	- (-) [0.97]	-0.486 (0.481) [0.97]	0.130 (0.186) [0.99]	-0.365 (0.567) [0.99]	-2.314 (2.160) [0.97]
Observations	4,993	4,352	3,512	2,607	1,897	1,213

Secondary / tertiary education. Raw point estimates are noisy and not concentrated at higher parities; no cell rejects under raw inference or under RW. Consistent with the lower-education subgroup carrying the higher-parity response.

Appendix: Turkish Mother Tongue

Outcome	Age 16	Age 20	Age 24	Age 28	Age 32	Age 36
<i>Turkish mother tongue</i>						
Ever Married	-0.433** (0.197) [0.37]	-1.286 (0.796) [0.53]	-0.143 (1.681) [0.98]	0.240 (1.621) [0.98]	-1.519 (1.295) [0.70]	-2.385 (2.949) [0.83]
Ever Birth ≥ 1	-0.172 (0.120) [0.94]	-0.813 (0.773) [0.98]	-0.569 (2.379) [1.00]	1.233 (1.928) [1.00]	-3.926 (2.778) [0.94]	-2.142 (3.325) [1.00]
Ever Birth ≥ 2	-0.003 (0.024) [1.00]	0.062 (0.747) [1.00]	-1.223 (0.906) [0.94]	0.191 (2.764) [1.00]	-0.779 (5.461) [1.00]	-3.663 (2.398) [0.92]
Ever Birth ≥ 3	0.004 (0.004) [0.99]	0.165 (0.317) [1.00]	-0.343 (0.570) [1.00]	-1.099 (1.785) [1.00]	2.768 (4.006) [1.00]	-0.849 (3.397) [1.00]
Ever Birth ≥ 4	- (-)	- (-)	-0.075 (0.325) [1.00]	-0.346 (1.064) [1.00]	2.582** (1.061) [0.70]	1.821 (2.959) [1.00]
Ever Birth ≥ 5	- (-)	- (-)	0.261** (0.115) [0.74]	-0.043 (0.310) [1.00]	0.026 (1.540) [1.00]	5.116 (3.823) [0.94]
Observations	11,013	10,512	9,655	8,137	6,634	4,960

Turkish mother-tongue women. Point estimates near zero across parities and ages; **no raw or RW rejection at higher parities**. The Kurdish/Arabic subgroup is the one that drives the higher-parity result.

Appendix: Rest of Turkey

Outcome	Age 16	Age 20	Age 24	Age 28	Age 32	Age 36
<i>Rest of Turkey</i>						
Ever Married	-0.261 (0.187) [0.54]	-1.358 (1.073) [0.56]	1.726 (1.784) [0.56]	0.047 (1.257) [0.97]	-2.056 (1.606) [0.56]	-1.905** (0.754) [0.10]
Ever Birth ≥ 1	0.045 (0.226) [1.00]	-0.394 (1.020) [1.00]	1.823 (1.400) [0.88]	1.304 (2.049) [0.99]	-6.070** (2.517) [0.56]	-0.989 (0.978) [0.96]
Ever Birth ≥ 2	-0.036 (0.043) [0.98]	-0.595 (0.616) [0.97]	-1.202 (0.967) [0.89]	2.665 (2.110) [0.89]	-4.895 (3.171) [0.80]	-1.143 (1.641) [0.99]
Ever Birth ≥ 3	-0.001 (0.001) [1.00]	0.185 (0.348) [1.00]	-0.654 (0.790) [0.98]	-0.135 (3.917) [1.00]	0.192 (3.875) [1.00]	0.594 (2.686) [1.00]
Ever Birth ≥ 4	- (-)	- (-)	0.227 (0.478) [1.00]	1.448 (1.377) [0.95]	1.494 (2.367) [0.99]	6.100** (2.545) [0.56]
Ever Birth ≥ 5	- (-)	- (-)	0.580** (0.270) [0.61]	0.892 (1.347) [0.99]	1.485 (1.636) [0.98]	5.437* (2.798) [0.65]
Observations	10,352	9,962	9,219	7,836	6,394	4,799

Provinces outside the East/Southeast high-exposure region. Point estimates small and mostly insignificant; the headline higher-parity response is **absent** away from the heavily-treated border region.

Appendix: Placebo with NUTS-1 Trends

Outcome	Age 16	Age 20	Age 24	Age 28	Age 32	Age 36
Ever Married	0.0004 (0.0005) [0.77]	0.0013 (0.0008) [0.54]	0.0016* (0.0010) [0.54]	0.0012** (0.0005) [0.38]	0.0014* (0.0008) [0.54]	-0.0004 (0.0007) [0.77]
Ever Gave Birth to One Child	0.0006** (0.0003) [0.72]	0.0014 (0.0010) [0.89]	0.0012 (0.0013) [0.98]	0.0013** (0.0006) [0.69]	0.0017** (0.0008) [0.72]	0.0002 (0.0007) [1.00]
Ever Gave Birth to Two Children	0.0001 (0.0002) [1.00]	-0.0000 (0.0005) [1.00]	0.0015* (0.0008) [0.76]	0.0025*** (0.0009) [0.54]	0.0024** (0.0011) [0.72]	0.0006 (0.0012) [1.00]
Ever Gave Birth to Three Children	0.0000 (0.0000) [0.94]	0.0001 (0.0003) [1.00]	0.0005 (0.0006) [0.99]	0.0002 (0.0007) [1.00]	-0.0000 (0.0007) [1.00]	-0.0024 (0.0014) [0.81]
Ever Gave Birth to Four Children	- (-)	- (-)	0.0005 (0.0004) [0.91]	0.0011 (0.0010) [0.94]	0.0009 (0.0024) [1.00]	-0.0012 (0.0028) [1.00]
Ever Gave Birth to Five Children	- (-)	- (-)	0.0000 (0.0002) [1.00]	0.0004 (0.0005) [0.99]	0.0013 (0.0015) [0.99]	0.0009 (0.0029) [1.00]
Observations	12,355	11,560	10,226	8,194	6,063	4,098

Same placebo specification as slide 9 ($Y \sim B_n \times t$ on $t(a, i) < 2012$) but with **NUTS-1-region** linear trends in place of macro-region trends. Coefficients remain near zero; **no RW rejection**.

Appendix: Placebo with NUTS-2 Trends

Outcome	Age 16	Age 20	Age 24	Age 28	Age 32	Age 36
Ever Married	-0.0004 (0.0005) [0.75]	0.0017 (0.0011) [0.45]	-0.0016 (0.0010) [0.44]	-0.0006 (0.0008) [0.75]	-0.0011 (0.0013) [0.75]	-0.0034** (0.0016) [0.26]
Ever Gave Birth to One Child	0.0010** (0.0004) [0.58]	0.0015 (0.0013) [0.95]	-0.0009 (0.0015) [0.99]	-0.0009 (0.0009) [0.95]	-0.0007 (0.0014) [0.99]	-0.0021 (0.0022) [0.95]
Ever Gave Birth to Two Children	0.0003 (0.0003) [0.95]	-0.0010 (0.0010) [0.95]	0.0010 (0.0015) [0.99]	0.0002 (0.0008) [0.99]	-0.0021* (0.0011) [0.73]	-0.0012 (0.0014) [0.97]
Ever Gave Birth to Three Children	0.0000 (0.0000) [0.99]	-0.0002 (0.0004) [0.99]	-0.0004 (0.0009) [0.99]	-0.0007 (0.0015) [0.99]	-0.0017 (0.0016) [0.95]	-0.0067* (0.0035) [0.69]
Ever Gave Birth to Four Children	- (-)	- (-)	0.0004 (0.0006) [0.99]	0.0022 (0.0019) [0.95]	0.0028 (0.0041) [0.99]	-0.0022 (0.0049) [0.99]
Ever Gave Birth to Five Children	- (-)	- (-)	-0.0001 (0.0003) [0.99]	0.0010 (0.0010) [0.95]	0.0037 (0.0024) [0.83]	0.0050 (0.0043) [0.94]
Observations	12,355	11,560	10,226	8,194	6,063	4,098

Same placebo specification as slide 9 but with **NUTS-2-region** (subregion) linear trends. Coefficients remain near zero across all cells; **no RW rejection** – pre-trend null is robust to finer trend structures.

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Appendix • Comparison with Aboulhosn, Aksoy & Özcan

Appendix: Comparison with Aboulhosn, Aksoy & Özcan (AAO)

Aboulhosn, Aksoy & Özcan: *Refugee Immigration and Natives' Fertility*.

- **Same research question:** how do Syrian refugees affect Turkish native fertility?
- **Different data:** they use the National Survey on Domestic Violence Against Women (**NSDVW**) 2008 and 2014; we use TDHS waves with a pseudo-panel structure.
- **Different design:** cross-sectional DiD on two waves vs. our continuous-DiD on an extended pseudo-panel.
- **Different outcome structure:** we track parity-by-age progression $\mathbb{1}\{\text{parity} \geq p \text{ by age } a\}$; they use contemporaneous 1-year birth/pregnancy indicators.

AAO headline IV results (Table 1).

Outcome	AAO IV
Gave birth past year	0.318**
Currently pregnant	0.248**
Number of children	1.870**
Birth & not 1st child	0.305***
Birth & 1st child	0.012
Pregnant & ≥ 1 child	0.317**

Reading they offer: *existing mothers*, not childless women, drive the response.

Appendix: AAO Table 1 Replication

Comparison on AAO's data (NSDVW): paper vs. our replication.

Outcome	Paper	Our replication
Gave birth past year	0.318**	-0.045
Currently pregnant	0.248**	0.612***
Number of children	1.870**	2.826***
Birth & not 1st child	0.305***	-0.045
Birth & 1st child	0.012	0.000
Pregnant & ≥ 1 child	0.317**	0.574***

Issues we identified in AAO's specification.

1. **Schooling controls:** asymmetric 2008/2014 recode; 2014 "never attended" $\Rightarrow$ schooling=0, 2008 "never attended" $\Rightarrow$ missing (737 women dropped).
2. **Dropping women with missing covariates:** 1,927 women dropped on missing birth-month/year despite valid age (miss count).
3. **Pregnancy definition:** inconsistent coding of missing across samples.
4. **No region time trends:** adding region $\times$ year FE flips the headline.

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[Next: AAO spec on TDHS](#)

Exercise. Re-run AAO's exact specification (same controls, same IV, same outcomes) on our **TDHS 2008 + 2013** ever-married 15–49 sample ($N = 14,621$).

Outcome	AAO paper IV	TDHS replication IV
Gave birth past year	0.318**	0.301
Currently pregnant	0.248**	−0.465**
Number of children	1.870**	2.588*
Birth & not 1st child	0.305***	−0.121
Birth & 1st child	0.012	0.421*
Pregnant & ≥ 1 child	0.317**	−0.223

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